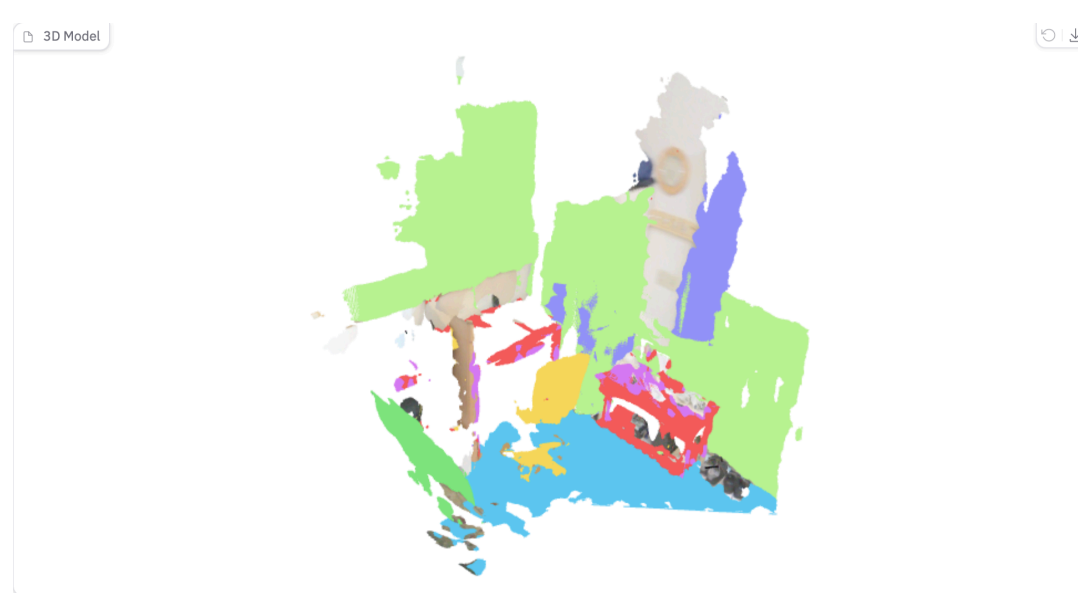


# Multi-View Consistent 3D Instance Segmentation on a Frozen VGGT Backbone

A D4RT-style / DETR-like decoder  
trained on SAM3 pseudo-labels  
(ScanNet)

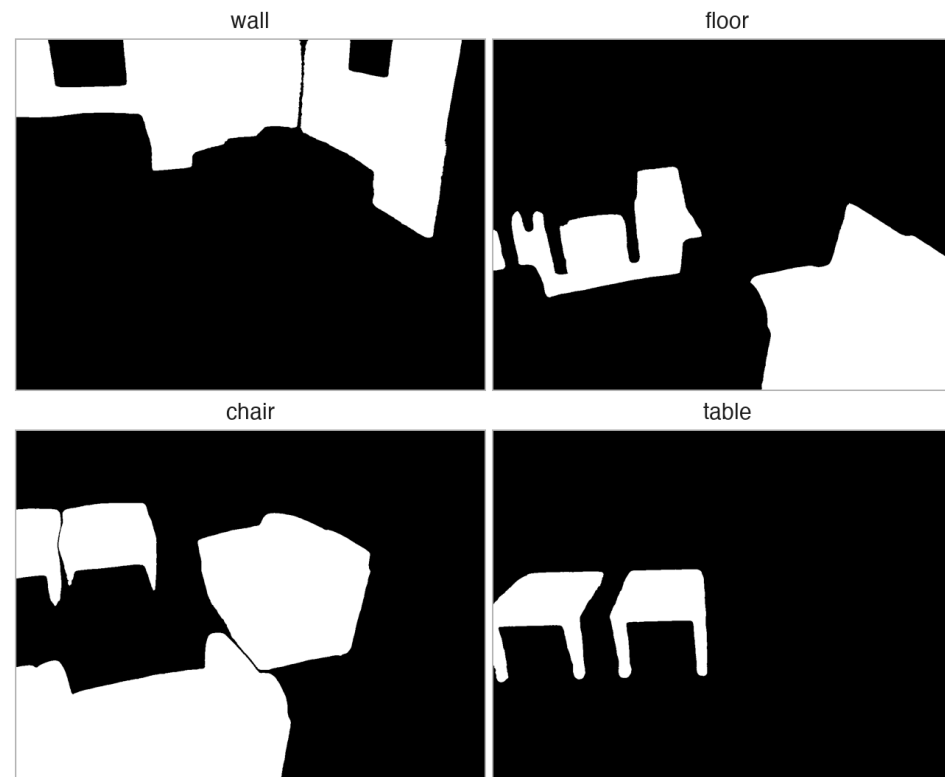
Nico Iacobone — Research update for supervision  
meeting

June 11, 2026



# 1 · Data & Supervision: ScanNet + SAM3 Pseudo-Labels

- For each scene a subset of ~100 images is selected; each frame is sampled 5 frames after the preceding one
- SAM3 is run on every frame, once per class: the result is one binary mask per class per frame (19 ScanNet classes, 0 is background)
- All masks of the same class are linked into a single instance, consistent across all views; each instance also receives a centroid point, used as its query prompt
- **Limitation:** binary per-class masks cannot separate two objects of the same class — a property of the data, not of the model



SAM3 per-class binary masks, one frame of `scene0001_00`

## 2 · The Decoder: Point-Prompted Queries → Multi-View Masks

The head reads a single tensor out of the frozen VGGT: the output of its last aggregator layer, i.e. the multi-view-fused features of all input frames. On top of it, the decoder turns *point prompts* into instance masks.

**A query is a point prompt:** a position  $(u, v)$  in one specific view. It is made of three pieces, summed into one vector:

- a **Fourier positional encoding of  $(u, v)$**  — tells the decoder *where* in the image the prompt sits
- a **learned view embedding** — tells it *which frame* the prompt belongs to; position alone is ambiguous when several views see the same scene
- a **small MLP on the  $9 \times 9$  RGB patch around the point** — tells it *what the point locally looks like*, so the query can bind to one specific object

**Instance decoder:** a standard 4-layer, 8-head transformer decoder. The VGGT features of all frames, projected to 256 dimensions and normalized, form the *memory*; the queries form the *target sequence* (tgt — the sequence the decoder refines layer by layer, alternating self-attention among queries and cross-attention into the memory). After decoding, the original query is added back to the output.

**Output heads:** a classification head (19 classes + background) and a mask head — each query produces a small mask kernel that is compared by cosine similarity with the per-pixel features of every frame. One query therefore emits one mask spanning all frames: multi-view consistency holds by construction.

### 3 · Matching & Losses

Training follows DETR-style set prediction: queries and ground-truth instances are matched one-to-one by the Hungarian algorithm, with a cost combining class probability, query-point distance and a dense mask cost. Each matched pair is then supervised with three losses:

- **Focal classification loss** — a cross-entropy that down-weights examples the model already gets right. Chosen because background queries vastly outnumber object queries: it keeps the few informative ones from being drowned out.
- **Dice mask loss** — measures the overlap between predicted and ground-truth mask, normalized by their sizes. Chosen because it is insensitive to the foreground/background pixel imbalance and optimizes the global shape of the mask, directly IoU-like.
- **Weighted binary cross-entropy mask loss** — per-pixel supervision with extra weight on foreground pixels. It complements Dice with precise, local gradients and prevents the trivial "everything is background" solution.

In a second phase of the project a *no-object* term was added: unmatched queries receive a small classification loss toward the background class. The model learns to answer "nothing here" — which is what makes inference without ground-truth prompts possible.

## 4 · Training Setup

Training the head is cheap: a full 1000-epoch run takes about 2 minutes on a single GPU.

- AdamW, learning rate  $2e-3$ , linear warmup (30 epochs) then cosine decay; gradient clipping
- **Regularization:** 3 frame bundles per scene (random frame re-sampling) · query jitter  $\sigma = 0.02$  · color jitter 0.2 · background queries resampled at every step
- **Model selection:** validation mIoU is measured every 50 epochs and the best checkpoint is kept; optional early stopping

## 5 · Quantitative Results

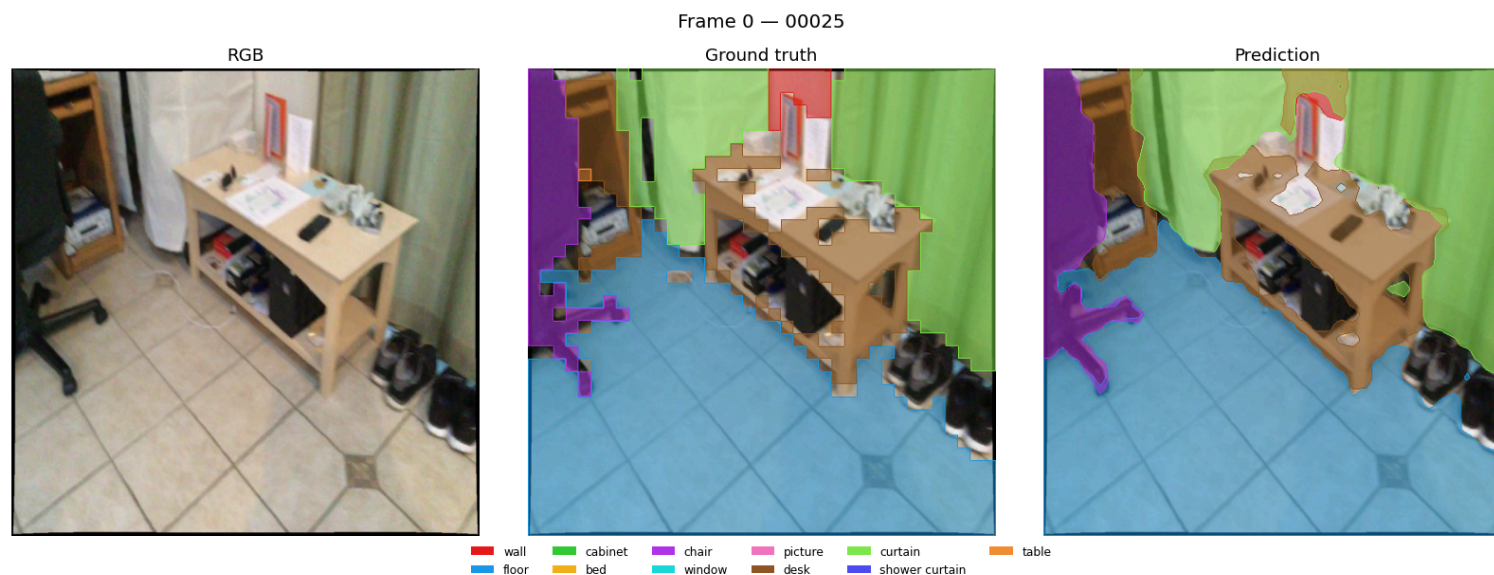
The pipeline demonstrably works (overfit ✓, multi-scene fit ✓, unprompted  $\approx$  prompted ✓); generalization is data-limited at 4 training scenes, exactly as expected.

| Experiment                                         | mIoU                             | AP50         | class_acc | Note                                                 |
|----------------------------------------------------|----------------------------------|--------------|-----------|------------------------------------------------------|
| Single-scene overfit (400 epochs)                  | 0.004 $\rightarrow$ <b>0.900</b> | 0.96         | 1.00      | gradient-flow sanity ✓                               |
| 4-scene training, fixed batch (mean)               | <b>0.967</b>                     | 0.54         | 0.94      | AP hurt by no bg supervision                         |
| — held-out scene (final ckpt)                      | 0.027                            | 0.00         | 0.29      | peaked $\sim$ 0.13 mid-training                      |
| 4-scene + regularization, prompted (mean)          | 0.666                            | <b>0.771</b> | 0.875     | AP50 0.54 $\rightarrow$ 0.77 via no-object loss      |
| 4-scene + regularization, <b>unprompted</b> (mean) | <b>0.678</b>                     | 0.446        | —         | $\approx$ prompted mIoU, <b>zero GT at test time</b> |
| — held-out scene, best ckpt                        | <b>0.138</b> @ ep 450            | —            | —         | $>$ 0.027 via model selection                        |

- **Prompted** = queries at GT centroids · **Unprompted** = uniform 6×6 grid per frame, background-argmax filtered — the honest detection number

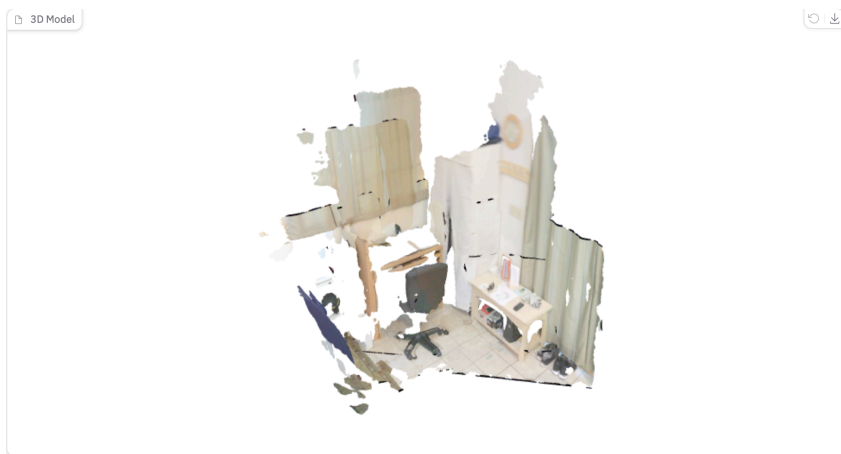
## 6 · Qualitative Results: 2D, Cross-View Consistent

Result after **overfitting on this single scene** and validating on the same scene (RGB | SAM3 ground truth | prediction). The same instance keeps the same color in every frame:

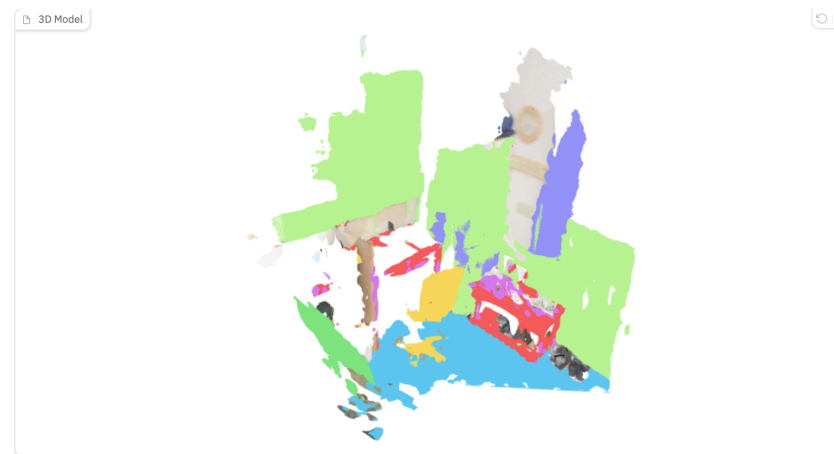


## 7 · Qualitative Results: 3D

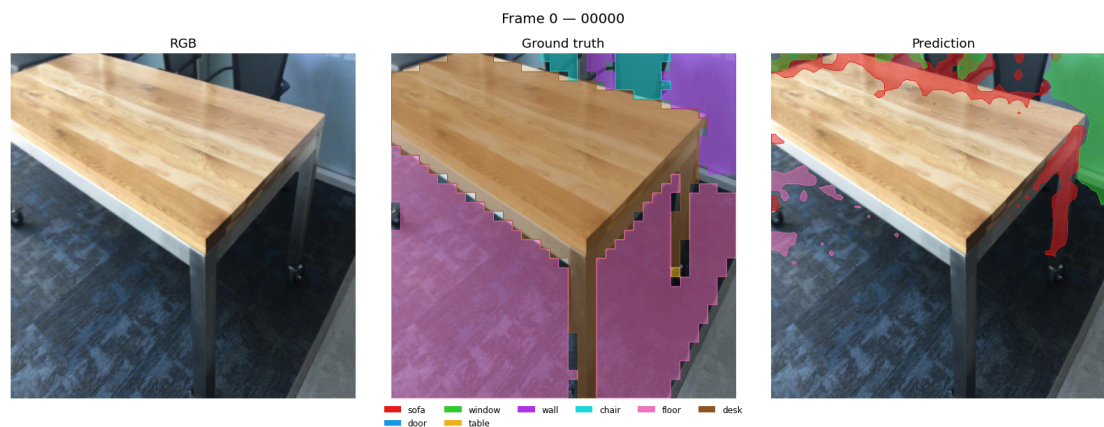
Model **overfitted on 4 scenes**; the bottom row shows the 5th, held-out validation scene.



VGGT reconstruction (RGB)



Same point cloud, colored by **predicted instance**



## 8 · Limitations, Open Questions & Next Steps

### Limitations

- Per-class binary SAM3 masks → can't separate same-class objects ("semantic-as-instance" ceiling)
- 37×37 mask resolution; no full-res upsampling yet
- Duplicate grid detections (no NMS) depress unprompted AP
- 5 scenes → no generalization claim yet

### Open questions

1. **Label format for the next SAM3 run:** semantic vs. instance vs. panoptic? — *blocks the preprocessing pipeline*
2. Point prompts vs. **learned object queries** — when to switch?

### Next steps

1. **Preprocess tens → hundreds of scenes** with SAM3 (after the label-format decision) — *the only blocker*
2. **Scaling experiment:**  $N \in \{10, 25, 50, 100+\}$ , held-out val, early stopping — *does held-out mIoU climb with N?* (N = 4: **0.138**)
3. Ablations once val signal > noise: no-object weight · augmentation · grid density (+ NMS)
4. Then: learned queries · mask upsampling · partial unfreezing